

Bzip2 is a compression format like ZIP and RAR, but it uses a different compression algorithm. Files compressed with bzip2 typically have the file extension .bz2, just as a file compressed with RAR has the .rar extension, or if we used ZIP (e.g., WinZIP or Windows' built-in compression), the file has the .zip extension. In Linux, the two most common formats are gzip (.gz) and bzip2 (.bz2); zip and rar are seldom used.

1. You have a file named dokument.tar.bz2 but cannot extract it with the command `bzip2 -d dokument.tar.bz2`, and you suspect the file is corrupted. Using the `man` command, you try to find a command that could potentially recover the file. Which command do you find, and how would you write it to attempt to recover the files in the corrupted archive dokument.tar.bz2?

First, I start my virtual machine then go to the terminal to find some instruction about bzip2 using the command `man`. When trying to do that it's come up with a message told me that command `bzip2` not found but can be installed with `/sudo apt install bzip2/`. I use this command `/sudo apt and install bzip2/` And after I put my password the process is beginning, and an installation is done after one or two minutes. Now I start again `man bzip 2` command and hit enter. It takes me to the full documentation of bzip2. Later I start reading the documentation until I come up with a section named recovering data from damage files. Here I found the command `bzip2recover` which can be used to recover damage files. Using this command would be like this `"bzip2recover dokument.tar.bz2"`. This command will generate `"recap_dear.x"` Where X is the number for every block that be recovered from the damage file in the same directory. Once the operation is done we can use this command `"bzip2 -d rec*file.bz2 > recovered_data"`.

```
vboxuser@Ubuntu24: ~  
No manual entry for bzip2  
vboxuser@Ubuntu24:~$ man bzip2  
No manual entry for bzip2  
vboxuser@Ubuntu24:~$ man bzip2  
No manual entry for bzip2  
vboxuser@Ubuntu24:~$ bzip2 --help  
Command 'bzip2' not found, but can be installed with:  
sudo apt install bzip2  
vboxuser@Ubuntu24:~$ sudo apt install bzip2  
[sudo] password for vboxuser:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Suggested packages:  
  bzip2-doc  
The following NEW packages will be installed:  
  bzip2  
0 upgraded, 1 newly installed, 0 to remove and 52 not upgraded.  
Need to get 34.0 kB of archives.  
After this operation, 189 kB of additional disk space will be used.  
Get:1 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 bzip2 arm64 1.0.8-5.1build0.1 [34.0 kB]  
]   
Fetched 34.0 kB in 0s (92.9 kB/s)  
Selecting previously unselected package bzip2.  
(Reading database ... 152202 files and directories currently installed.)  
Preparing to unpack .../bzip2_1.0.8-5.1build0.1_arm64.deb ...  
Unpacking bzip2 (1.0.8-5.1build0.1) ...  
Setting up bzip2 (1.0.8-5.1build0.1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
vboxuser@Ubuntu24:~$
```

```
vboxuser@Ubuntu24: ~  
-8      6800k    3300k    2100k    828642  
-9      7600k    3700k    2350k    828642  
  
RECOVERING DATA FROM DAMAGED FILES  
  
  bzip2 compresses files in blocks, usually 900 kbytes long. Each block is handled independently. If a media or transmission error causes a multi-block .bz2 file to become damaged, it may be possible to recover data from the undamaged blocks in the file.  
  
  The compressed representation of each block is delimited by a 48-bit pattern, which makes it possible to find the block boundaries with reasonable certainty. Each block also carries its own 32-bit CRC, so damaged blocks can be distinguished from undamaged ones.  
  
  bzip2recover is a simple program whose purpose is to search for blocks in .bz2 files, and write each block out into its own .bz2 file. You can then use bzip2 -t to test the integrity of the resulting files, and decompress those which are undamaged.  
  
  bzip2recover takes a single argument, the name of the damaged file, and writes a number of files "rec00001file.bz2", "rec00002file.bz2", etc., containing the extracted blocks. The output filenames are designed so that the use of wildcards in subsequent processing -- for example, "bzip2 -dc rec*file.bz2 > recovered_data" -- processes the files in the correct order.  
  
  bzip2recover should be of most use dealing with large .bz2 files, as these will contain many blocks. It is clearly futile to use it on damaged single-block files, since a damaged block cannot be recovered. If you wish to minimise any potential data loss through media or transmission errors, you might consider compressing with a smaller block size.
```

2. In which section of the manual pages can you find manual pages for games?

I can find the manual page for games in section 6 by using the command `man man`.

3. When you have opened a manual page and are reading it, how do you search within the page to quickly find the right information? Which command do you use?

And reading the documentation about the manual using `man man`, I found a command "`man -k keyword`". We can use this command and then instead of keywords we can use anything we want to search for, and it will come with all results in the hold documentation over the manual.